SMO-Diff: A Morphology-Oriented Diffusion Model for Image-Level Outcome Prediction and Parameter Estimation in Scoliosis Surgery

***Abstract*— Precise individualized preoperative planning is essential for deformity correction in adolescent idiopathic scoliosis (AIS). Fusion levels, pedicle screw distribution, and rod configuration directly affect postoperative alignment, biomechanical stability, and long-term outcomes. Current planning relies heavily on manual measurement and surgeon experience, which is time-consuming and subjective. We propose SMO-Diff, a spinal morphology-oriented conditional diffusion framework for jointly predicting postoperative radiographs and internal-fixation masks from preoperative spinal radiographs. SMO-Diff first constructs a postoperative spinal centerline as a global morphological constraint, then uses a Curve Encoder and a region-of-interest (ROI) Encoder to extract global curve features and local surgical-region features. A Curve-Morphology Fusion (CMF) module integrates these complementary priors and injects multi-scale structural guidance into the diffusion generation network. The model synthesizes anatomically consistent postoperative radiographs and outputs screw and rod masks for interpretable internal-fixation planning. Experimental results show that SMO-Diff improves image fidelity, anatomical alignment, and mask quality, enabling quantitative evaluation of key surgical planning parameters and providing a reproducible visual aid for intelligent preoperative planning in scoliosis surgery.**

***Index Terms*— Adolescent Idiopathic Scoliosis (AIS), spinal morphology, conditional diffusion model, scoliosis surgery, postoperative radiographic prediction, preoperative planning**

I. Introduction

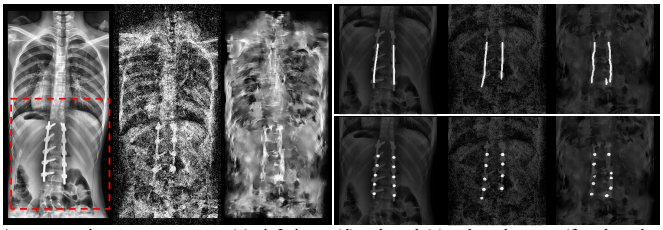
ADOLESCENT idiopathic scoliosis (AIS) is the most common structural spinal deformity in adolescents [1] and remains a major indication for corrective spinal surgery [2]. For patients requiring operative treatment, accurate preoperative planning is essential because fusion levels, screw distribution, and rod configuration must be tailored to the patient-specific deformity pattern and spinal morphology. [3] These decisions directly affect postoperative alignment, biomechanical stability, and long-term outcomes. [4] In current clinical practice, however, surgical planning still relies heavily on manual radiographic measurements, deformity classification, and surgeon experience [5], making the process labor-intensive, subjective, and difficult to standardize. An automated framework that can infer key surgical planning parameters from preoperative radiographs would therefore provide an objective efficient, and reproducible tool for individualized AIS surgery planning.

Despite its clinical significance, automatic surgical parameter planning for AIS is intrinsically difficult. Surgical decisions are not determined by a single radiographic parameter, but by the coupled relationships among curve type, deformity severity, spinal alignment, compensatory segments, and the anticipated correction strategy. [6] In particular, parameters such as the upper and lower instrumented vertebrae, screw density, correction extent, and rod contour are strongly interdependent rather than independently assigned. [7] A clinically useful model must therefore move beyond isolated measurement or classification and instead perform structured reasoning over spinal morphology, surgical-region localization, and multiple coordinated planning variables under anatomical and biomechanical constraints.

Deep learning has already shown considerable promise in AIS-related imaging and decision support. Existing studies have mainly focused on automated quantitative analysis of preoperative images, including Cobb angle measurement [8], biplanar alignment assessment [9], and Lenke classification [10], while predictive models have largely addressed curve progression [11] and related disease-management tasks. More recently, a small number of studies have begun to explore surgery-related applications. For example, pedicle screw selection [12], and placement [13], trajectory planning [14], and rod material choice [15]. These studies suggest that deep learning is beginning to shift AIS research from image analysis toward treatment-oriented prediction.

However, current work remains limited in a clinically important sense. Most existing studies either focus on a single planning-related variable or predict outcomes that are not directly actionable for preoperative surgical planning, such as progression risk [16], postoperative coronal correction [12], or rehabilitation-related outcomes [17]. Even the few surgery-oriented models reported to date cover only a limited subset of planning variables and fail to provide a systematic set of mutually related parameters for individualized operative planning [18]. More importantly, existing methods generally do not generate postoperative fixation layouts at the image level. Consequently, they provide only numerical predictions of isolated planning variables rather than a coherent surgical plan, limiting their usefulness for individualized operative decision making. Consequently, it remains an unresolved challenge to predict multiple parameters directly relevant to surgical planning from preoperative radiographs while simultaneously providing an intuitive representation of the expected postoperative fixation outcome.

Conditional generative modeling offers a promising direction for addressing this gap. However, existing methods remain inadequate for tasks that require precise anatomical constraints and explicit surgical logic. As illustrated in Fig. 1, mainstream generative models, including pix2pix [19] under the GAN architecture and Fast-DDPM [20] under the diffusion architecture, fail to produce clinically practical results for our task of joint postoperative radiograph synthesis and internal-fixation mask prediction. They typically generate noisy or blurry spinal tissues, discontinuous rod contours, and poorly aligned screw positions, lacking the anatomical accuracy and structural consistency required to reflect real surgical outcomes. These defects prevent reliable reconstruction of spinal topology and precise spatial correspondence of internal fixation, limiting their utility for individualized scoliosis surgery planning. More broadly, most generative frameworks are designed for general image-to-image translation rather than spinal surgery, and are therefore ill-suited to jointly model global spinal morphology, screw distribution, and rod configuration.



(a) postoperative radiograph (GT) (b) Fast-DDPM (c) pix2pix (d) rods and screws(GT) (e) rods and screws (Fast-DDPM) (f) rods and screws(pix2pix)

Fig. 1. Representative performance of existing generative models for postoperative radiograph synthesis and internal-fixation reconstruction. (a) GT radiograph (red dashed box marks the surgical region). (b) Fast-DDPM [20] produces severe noise. (c) pix2pix [19] generates blurry anatomy. (d) GT rod and screw masks. (e) Fast-DDPM fails to reconstruct continuous rods. (f) pix2pix yields discontinuous rods and misaligned screws.

To address these limitations, in this work, we propose a **S**pinal **M**orphology-**O**riented conditional **Diff**usion framework (entitled SMO-Diff) for joint prediction of postoperative fixation outcomes and surgical planning parameters from preoperative spinal radiographs. SMO-Diff first constructs a postoperative spinal centerline to provide global morphological guidance. A Curve Encoder and a region-of-interest (ROI) Encoder then extract complementary priors from the centerline and surgically relevant regions, respectively, while a Curve-Morphology Fusion (CMF) module integrates these features and injects multi-scale structural guidance into the diffusion process. In this way, the framework can synthesize anatomically consistent postoperative images while jointly predicting screw and rod masks together with associated planning parameters. By linking parameter-level prediction with image-level fixation-layout estimation, SMO-Diff provides a more intuitive and practically useful tool for individualized scoliosis surgery planning.

The main contributions of this work are threefold:

- First, we propose a morphology-oriented conditional diffusion framework for joint postoperative radiograph synthesis and internal-fixation mask prediction in AIS surgical planning.
- Second, we introduce a Curve Encoder, a ROI Encoder, and a CMF module that integrates global spinal alignment priors with local surgical-region constraints.
- Third, we collect a dedicated clinical AIS dataset and conduct extensive experiments demonstrating that the proposed method achieves superior performance compared with several advanced and mainstream generative models.

II. Related Works

A. AI-Based Analysis and Surgical Planning for Adolescent Idiopathic Scoliosis

Artificial intelligence (AI) has increasingly been applied to the diagnosis, assessment, and management of adolescent idiopathic scoliosis (AIS). Early studies mainly focused on automating radiographic measurements, particularly Cobb angle estimation. Zhang et al. proposed an automatic Cobb angle measurement method based on fuzzy Hough transform with vertebral shape priors, demonstrating the feasibility of computer-assisted scoliosis assessment [22]. Subsequently, deep learning-based approaches have improved the robustness and clinical applicability of automated measurement. Sun et al. developed a vertebral keypoint detection framework for Cobb angle measurement and reported comparable performance between automated and manual assessment [23]. Wong et al. further validated an artificial intelligence-based Cobb angle measurement system on pediatric AIS radiographs, supporting its potential clinical application [24]. Beyond Cobb angle measurement, recent studies have extended AI methods to automated assessment of spinal alignment parameters [25], vertebral landmark detection [23], and Lenke classification of AIS curves [26], [27].

In addition to diagnostic assessment, machine learning models have been explored for predicting scoliosis progression and surgical outcomes. Wang et al. developed a deep learning model using spinal radiographs obtained at the first clinic visit to predict AIS progression risk [28]. A recent systematic review summarized the application of machine learning methods for AIS progression prediction and highlighted the growing role of AI-assisted clinical decision support [11]. Furthermore, Abedi et al. investigated postoperative outcome prediction after scoliosis correction surgery using an adaptive neuro-fuzzy inference system [29]. These studies demonstrate the potential of AI in supporting quantitative surgical decision-making; however, most existing approaches remain limited to predicting individual parameters or specific clinical indices rather than modeling the comprehensive postoperative surgical result.

AIS corrective surgery is a complex procedure influenced by multiple interacting factors, including spinal morphology, fusion levels, implant configuration, correction strategies, and postoperative alignment. Recent work such as ScoliosisPLAN has explored AI-assisted scoliosis surgical planning and postoperative prediction [30]; however, existing intelligent systems still mainly focus on numerical prediction of selected surgical variables. The generation of visually interpretable postopera-

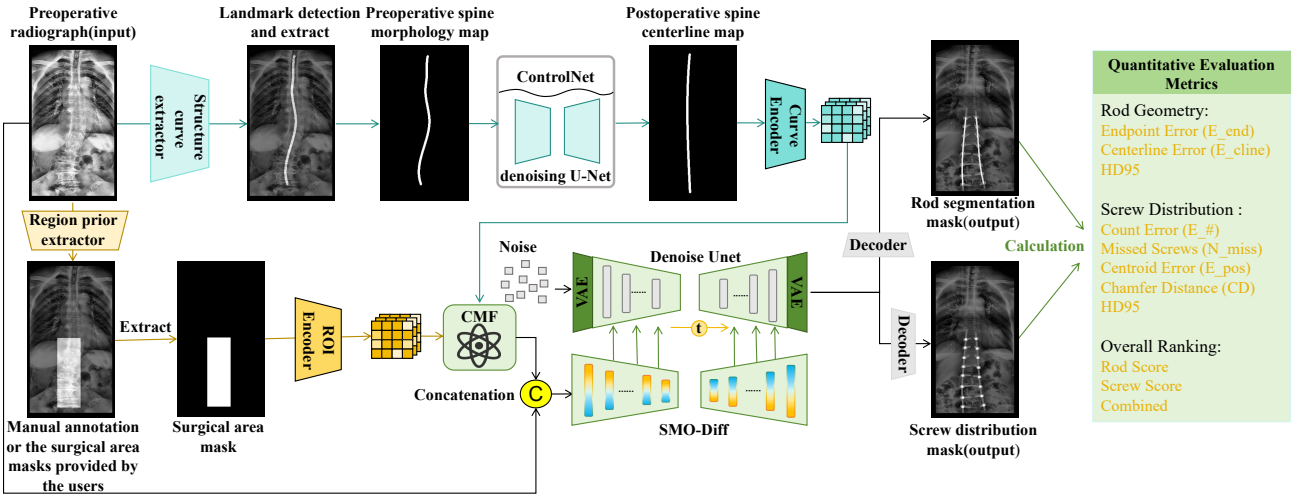


Fig. 2. Overview of the proposed SMO-Diff framework. The preoperative spinal centerline is first extracted and then fed into ControlNet [21] to predict the postoperative centerline as a global morphological constraint. Together with the preoperative radiograph and the surgical region of interest, the predicted centerline guides the diffusion network to synthesize the postoperative radiograph and output the corresponding screw and rod masks. Specifically, the predicted postoperative centerline and the surgical region are encoded by the Curve Encoder and the ROI Encoder, respectively; their complementary priors are then integrated via a CMF module to inject multi-scale structural guidance into the denoising network (see Fig. 4 for CMF details).

tive outcomes, including corrected spinal morphology and implant configuration, remains insufficiently explored. Therefore, an intelligent framework capable of jointly modeling postoperative radiographic appearance and implant-related structures is needed to provide more comprehensive and intuitive assistance for individualized surgical planning.

B. Implant-Related Planning and Fixation Layout Prediction in Scoliosis Surgery

Implant configuration is a critical component of AIS corrective surgery because the distribution of pedicle screws and rod contouring directly influences correction performance, biomechanical behavior, and postoperative spinal alignment. Bharucha et al. investigated low-density versus high-density thoracic pedicle screw constructs and evaluated their effects on surgical outcomes in AIS correction [31]. Shen et al. further compared different pedicle screw densities in Lenke 1 AIS patients and analyzed their relationship with correction outcomes [32]. From a biomechanical perspective, Clin et al. compared the load-sharing capacity of high- and low-density implant constructs with different pedicle screw configurations, demonstrating the importance of implant density in spinal instrumentation mechanics [33]. In addition, Wang et al. investigated how differential rod contouring contributes to three-dimensional correction and affects bone-screw forces during AIS instrumentation [34].

Traditional implant planning relies heavily on surgeon experience, radiographic measurements, and biomechanical simulation. Although these approaches provide valuable information regarding fusion levels, screw placement, and correction strategies, they generally require manually defined parameters and do not automatically infer patient-specific fixation layouts from preoperative imaging.

Recent AI-based studies have improved automation in scoliosis assessment, including vertebral landmark detection [23], radiographic parameter measurement [24], [25], and surgical parameter prediction [30]. However, most existing methods formulate surgical planning as a parameter estimation problem rather than a direct prediction of postoperative imaging outcomes. Consequently, the spatial relationship among corrected spinal anatomy, screw distribution, and rod geometry is not explicitly modeled. Developing an image-level implant-aware prediction framework may provide a more interpretable representation of surgical outcomes and facilitate personalized AIS surgical planning.

C. Generative Models for Medical Image Prediction and Surgical Outcome Simulation

Generative models have recently emerged as powerful approaches for medical image synthesis due to their ability to learn complex anatomical distributions and generate realistic imaging data. GAN-based methods, particularly pix2pix, have been widely adopted for image-to-image translation tasks by learning mappings between different imaging domains [19]. In medical imaging, MedGAN demonstrated the potential of GAN-based frameworks for medical image translation and synthesis [35], while other studies explored GAN-based cross-contrast image generation and reconstruction tasks [36].

More recently, diffusion models have achieved significant progress in medical image generation due to their stable optimization and high-quality synthesis capability. The denoising diffusion probabilistic model (DDPM) introduced a probabilistic framework for high-fidelity image generation [37]. Building upon diffusion models, ControlNet incorporated spatial conditions such as edges, contours, and segmentation maps to enable controllable image synthesis [21]. In medical imaging, Konz et al. proposed a segmentation-guided diffusion framework for anatomically controllable medical image generation, demonstrating that anatomical constraints improve structural consistency and generation reliability [38].

Beyond general medical image generation, generative models have also been investigated in scoliosis-related imaging. He et al. developed a conditional generative adversarial network-based framework to generate virtual spinal radiographs from smartphone photographs, enabling radiation-free scoliosis evaluation and Cobb angle assessment [39]. Zhou et al. proposed UXDiff, a diffusion-based method for translating coronal ultrasound spine images into radiograph-like images to improve visualization of spinal structures [40]. Although these studies demonstrate the feasibility of generative models in scoliosis imaging, they mainly focus on modality translation, screening, or visualization enhancement rather than surgery-induced anatomical transformation.

Postoperative scoliosis simulation requires modeling complex changes including spinal curvature correction, vertebral realignment, global balance restoration, and implant configuration. Recent AI-assisted surgical planning studies have begun to explore postoperative prediction [30]; however, generation of postoperative radiographs together with explicit implant representations such as pedicle screw and rod layouts remains largely unexplored. Therefore, a specialized generative framework integrating anatomical priors, implant constraints, and surgical correction patterns is needed for personalized AIS surgical simulation and outcome prediction.

III. Method

A. Problem Formulation

This work considers the surgical planning as a spinal structure-guided surgical parameter prediction problem. Given a preoperative radiograph X_{pre}, its preoperative spinal centerline C_{pre} and the planned surgical region R, the objective is to predict surgery-related parameters (e.g., screw distribution and rod configuration) under anatomical and structural constraints. To achieve this goal, SMO-Diff reformulates surgical parameter prediction as an image-level postoperative fixation outcome prediction task. Specifically, SMO-Diff first predicts the postoperative radiographic representation and the corresponding fixation masks:

$$\left(\hat{X}_{\mathrm{post}}, \hat{M}_{\mathrm{s}}, \hat{M}_{\mathrm{r}},\right) = G_\theta\left(I_{\mathrm{pre}}, C_{\mathrm{pre}}, R\right), \tag{1}$$

where G_θ denotes the proposed SMO-Diff, $\hat{X}_{\mathrm{post}}$ is the predicted postoperative radiographic representation, and $\hat{M}_{\mathrm{s}}$ and $\hat{M}_{\mathrm{r}}$ denote the predicted screw and rod masks, respectively. The surgery-related parameters are then estimated from the predicted fixation outcomes:

$$\hat{\mathbf{p}} = Q_\omega\left(\hat{X}_{\mathrm{post}}, \hat{M}_{\mathrm{s}}, \hat{M}_{\mathrm{r}}\right), \tag{2}$$

where Q_ω dentoes the parameter estimation module and $\hat{\mathbf{p}}$ represents the predicted surgical planning parameters. In this formulation, surgical parameter estimation is constrained by postoperative anatomical and fixation patterns, allowing the model to produce more structurally consistent and clinically interpretable planning results.

B. Postoperative Spinal Centerline Prediction

Given the preoperative spinal centerline C_{pre}, SMO-Diff first estimate the postoperative centerline using a ControlNet module curve prediction network [21]. The predicted postoperative centerline is:

$$C_{\mathrm{post}} = F_{\mathrm{g}}(C_{\mathrm{pre}})\,. \tag{3}$$

This formulation models the geometric deformation of the spine independently of grayscale intensities in radiographs, yielding a clear structural prior C_{post} to regularize the subsequent radiograph synthesis.

C. Postoperative Radiograph Generation

SMO-Diff generates the postoperative radiograph from the preoperative radiograph X_{pre}, postoperative centerline C_{post}, and surgical region R, where R denotes the surgical region annotated by clinicians or derived from preoperative planning. The target is the predicted postoperative radiograph $\widehat{X}_{\mathrm{post}}$.

During training, the ground-truth postoperative radiograph X_{post} is first encoded into the latent space as $z_0 = E_{\mathrm{VAE}}(X_{\mathrm{post}})$. Then, Gaussian noise is added at diffusion step t according to

$$z_t = \sqrt{\alpha_t} z_0 + \sqrt{1-\alpha_t}\epsilon, \tag{4}$$

where $\epsilon \sim \mathcal{N}(0, I)$ and α_t is the noise-scheduling coefficient.

To guide synthesis, SMO-Diff constructs conditional features from preoperative appearance, postoperative morphology, and the surgical region. Specifically, the preoperative image is encoded as $F_x = E_x(X_{\mathrm{pre}})$, while the postoperative centerline and surgical region are encoded as $F_c = E_c(C_{\mathrm{post}})$ and $F_r = E_r(R)$, respectively.

The CMF module fuses the structural features to obtain $F_{cr} = \mathcal{F}_{\mathrm{fusion}}(F_c, F_r)$. The fused structural prior is then concatenated with the preoperative appearance feature, yielding the final condition $F_{\mathrm{cond}} = \mathrm{Concat}(F_x, F_{cr})$. Based on this condition, multi-scale guidance features $\{S^1, S^2, \ldots, S^L\}$ are extracted by $\mathcal{F}_{\mathrm{ms}}(\cdot)$ and injected into different denoising layers.

The denoising network predicts the noise component as

$$\widehat{e} = e_\theta(z_t, t, \{S^1, S^2, \ldots, S^L\}), \tag{5}$$

and is optimized using the diffusion loss

$$\mathcal{L}_{\mathrm{diff}} = \|e - \widehat{e}\|_2^2. \tag{6}$$

After reverse diffusion, the estimated latent representation $\widehat{z}_0 = \mathcal{D}_{\mathrm{reverse}}(z_t, \widehat{e}, t)$ is decoded by the VAE decoder to obtain the predicted postoperative radiograph, $\widehat{X}_{\mathrm{post}} = D_{\mathrm{VAE}}(\widehat{z}_0)$.

Finally, a lightweight fixation-visualization module predicts internal-fixation maps from the generated postoperative radiograph:

$$\widehat{M} = G_\phi(\widehat{X}_{\mathrm{post}}), \qquad \widehat{M} = [\widehat{M}_{\mathrm{screw}}, \widehat{M}_{\mathrm{rod}}], \tag{7}$$

where $\widehat{M}_{\mathrm{screw}}$ and $\widehat{M}_{\mathrm{rod}}$ are binary masks for screw and rod localization, enabling direct visualization of the fixation layout.

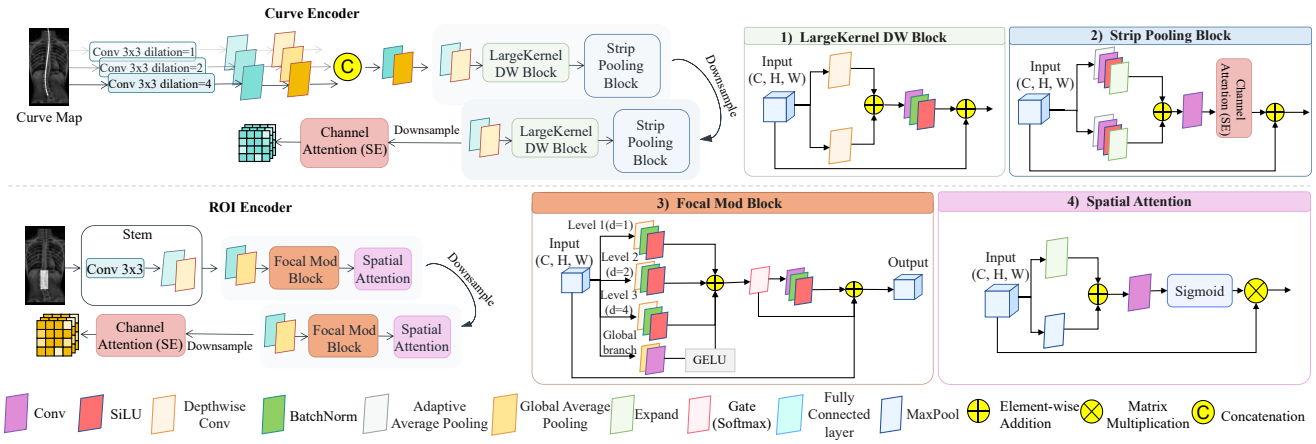


Fig. 3. Overview of the proposed Curve Encoder and ROI Encoder modules. The channel attention (SE) blocks implement the squeeze-and-excitation mechanism to adaptively recalibrate channel-wise feature responses, thereby emphasizing informative anatomical channels while suppressing less relevant background responses [41].

1) Curve Encoder for Postoperative Spinal Centerline Encoding: The postoperative centerline map $C_{\text{post}} \in \mathbb{R}^{1\times H\times W}$ is a sparse single-channel representation describing the spinal centerline after surgery. The Curve Encoder extracts global spinal alignment, long-range continuity, and local curvature variation, producing a curve-aware representation:

$$F_c = \mathcal{E}_{\text{curve}}(C_{\text{post}}) \in \mathbb{R}^{64\times H/4\times W/4}. \tag{8}$$

As shown in Fig. 3, the input centerline map is first processed by three parallel dilated convolution branches with dilation rates 1, 2, and 4:

$$F_0 = \mathcal{P}_{24}\left(\text{Concat}\left(\mathcal{D}_1(C_{\text{post}}), \mathcal{D}_2(C_{\text{post}}), \mathcal{D}_4(C_{\text{post}})\right)\right). \tag{9}$$

The Curve Encoder then progressively extracts curve-aware features through two hierarchical stages:

$$F_1 = \mathcal{S}_{32}\left(\mathcal{L}_{32}\left(\mathcal{P}_{32}(F_0)\right)\right), \tag{10}$$

$$F_2 = \mathcal{S}_{64}\left(\mathcal{L}_{64}\left(\mathcal{P}_{64}\left(\mathcal{D}_{\text{down}}(F_1)\right)\right)\right), \tag{11}$$

$$F_c = \mathcal{C}_{\text{out}}\left(\mathcal{D}_{\text{down}}(F_2)\right). \tag{12}$$

Here, $\mathcal{D}_d(\cdot)$ denotes a dilated convolution branch with dilation rate d, $\mathcal{P}(\cdot)$ denotes a 1×1 projection layer, $\mathcal{L}(\cdot)$ denotes the LargeKernel DW Block, $\mathcal{S}(\cdot)$ denotes the Strip Pooling Block, $\mathcal{D}_{\text{down}}(\cdot)$ denotes stride-2 downsampling, and $\mathcal{C}_{\text{out}}(\cdot)$ denotes the output channel attention module.

Thus, F_c encodes both global spinal alignment and local curvature structures, providing structural guidance for anatomically consistent generation.

2) ROI Encoder for Surgical Region Encoding: The surgical-region prior $R \in \mathbb{R}^{1\times H\times W}$ describes the estimated operative area from clinical experience or preoperative planning, providing local constraints for implant placement and tissue deformation. The ROI Encoder maps this prior to a compact local constraint representation:

$$F_r = \mathcal{E}_{\text{ROI}}(R) \in \mathbb{R}^{64\times H/4\times W/4}. \tag{13}$$

As shown in Fig. 3, the input prior is first processed by a convolutional stem:

$$G_0 = \mathcal{T}_{\text{stem}}(R) \in \mathbb{R}^{24\times H\times W}, \tag{14}$$

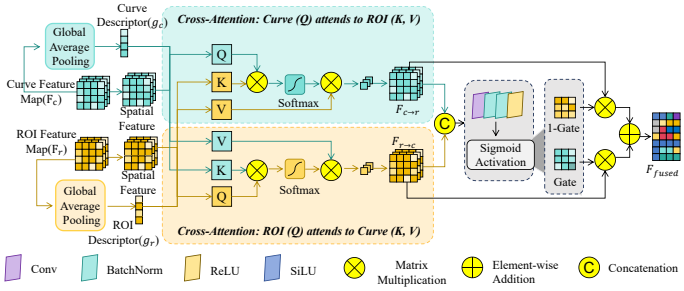


Fig. 4. Overview of the proposed CMF module.

where $\mathcal{T}_{\text{stem}}(\cdot)$ consists of a 3×3 convolution, normalization, and a 1×1 projection.

The ROI Encoder then progressively extracts local surgical-region features through two hierarchical stages:

$$G_1 = \mathcal{A}_{\text{sp}}^{32}\left(\mathcal{M}_{\text{focal}}^{32}\left(\mathcal{P}_{32}(G_0)\right)\right), \tag{15}$$

$$G_2 = \mathcal{A}_{\text{sp}}^{64}\left(\mathcal{M}_{\text{focal}}^{64}\left(\mathcal{P}_{64}\left(\mathcal{D}_{\text{down}}(G_1)\right)\right)\right), \tag{16}$$

$$F_r = \mathcal{C}_{\text{out}}\left(\mathcal{D}_{\text{down}}(G_2)\right). \tag{17}$$

Here, $\mathcal{P}_{32}(\cdot)$ and $\mathcal{P}_{64}(\cdot)$ denote 1×1 projection layers, $\mathcal{M}_{\text{focal}}(\cdot)$ denotes the Focal Modulation Block, $\mathcal{A}_{\text{sp}}(\cdot)$ denotes the Spatial Attention Block, $\mathcal{D}_{\text{down}}(\cdot)$ denotes stride-2 downsampling, and $\mathcal{C}_{\text{out}}(\cdot)$ denotes the output channel attention module.

Thus, F_r encodes the location, extent, and local structural properties of the surgical region, providing spatial constraints for anatomically plausible implant generation and tissue deformation.

3) CMF for Curve–ROI Interaction: The proposed Curve-Morphology Fusion (CMF) module explicitly models the interaction between the global curve representation and the local ROI representation. Given the outputs of the Curve Encoder and ROI Encoder:

$$F_c, F_r \in \mathbb{R}^{C\times H'\times W'}, \tag{18}$$

where $C = 64$, $H' = H/4$, and $W' = W/4$, we first obtain their spatial tokens and global descriptors:

$$X_c = \phi(F_c), \qquad X_r = \phi(F_r), \tag{19}$$

$$g_c = \mathrm{GAP}(F_c), \qquad g_r = \mathrm{GAP}(F_r), \tag{20}$$

where $\phi(\cdot)$ denotes spatial flattening, $X_c, X_r \in \mathbb{R}^{N\times C}$, and $N = H'W'$.

As shown in Fig. 4, CMF contains two bidirectional cross-attention branches. We denote the cross-attention operation from feature X_b to feature X_a as:

$$\mathcal{X}_{a\to b} = \phi^{-1}\left(\mathrm{Softmax}\left(\frac{Q_a K_b^\top}{\sqrt{d}}\right) V_b\right), \tag{21}$$

where Q_a, K_b, and V_b are generated inside the cross-attention branch, as illustrated in Fig. 4.

The bidirectional curve–ROI interaction is then formulated as:

$$F_{c\to r} = \mathcal{X}_{c\to r}\left(X_c, X_r; g_c, g_r\right), \tag{22}$$

$$F_{r\to c} = \mathcal{X}_{r\to c}\left(X_r, X_c; g_r, g_c\right), \tag{23}$$

where $F_{c\to r}$ denotes curve features enhanced by ROI constraints, and $F_{r\to c}$ denotes ROI features guided by global curve alignment.

The two cross-modal features are concatenated and passed through a lightweight gating branch:

$$F_{\mathrm{cat}} = \mathrm{Concat}\left(F_{c\to r}, F_{r\to c}\right), \tag{24}$$

$$G = \mathcal{G}\left(F_{\mathrm{cat}}\right), \qquad G \in \mathbb{R}^{C\times H'\times W'}, \tag{25}$$

where $\mathcal{G}(\cdot)$ denotes the gating branch shown in Fig. 4.

Finally, the fused representation is obtained by gated element-wise aggregation:

$$F_{\mathrm{fused}} = (1 - G)\odot F_{c\to r} + G \odot F_{r\to c}, \tag{26}$$

where $\odot$ denotes element-wise multiplication. The resulting feature F_{fused} is injected into the conditional branch of the denoising U-Net to guide anatomically consistent generation.

IV. Experiments and Results

A. Dataset and Data Preprocessing

This retrospective study enrolled 612 eligible AIS patients (age 10–18 years) who underwent posterior corrective fusion surgery at Queen Mary Hospital between June 2023 and August 2025. The study was approved by the HKU/HA HKW IRB (No. UW 25-300, July 8, 2025), with informed consent obtained from all patients and/or guardians. Inclusion criteria required a clinical and radiographic diagnosis of AIS, complete surgical records, and available preoperative and postoperative spinal AP/Lat radiographs together with surgical planning data. Exclusion criteria comprised other spinal deformities, prior spinal surgery, or poor imaging quality.

Raw DICOM images were converted to 8-bit grayscale, resized to 240×480 pixels, and preprocessed using adaptive contrast enhancement and Gaussian denoising ($\sigma = 1.0$, 3×3 kernel). Preoperative spinal ROI masks and vertebral corner landmarks were manually annotated by two senior spine surgeons in a double-blinded manner, with disagreements resolved through consensus. For each vertebra, four anatomical corner points were labeled, and the vertebral center was determined by the intersection of the two diagonals. A smooth spinal centerline was then constructed by connecting these centers, serving as a compact structural prior of global spinal alignment and curvature.

The dataset was partitioned into training, validation, and test sets at an 8:1:1 ratio. All experiments were conducted on a single NVIDIA GeForce RTX 3090 (24 GB) GPU. Models were implemented in PyTorch and PyTorch Lightning with batch size 8 and the Adam optimizer. Training was terminated upon convergence without observed overfitting.

B. Experiment Setup and Evaluation Metrics

For each predicted mask and GT mask, we extract connected components for screws and rods separately. Let Ω denote the two-dimensional image domain. For the ith test case and method a, the predicted and ground-truth screw masks are denoted by S_i^a and S_i^{gt}, and the predicted and GT rod masks are denoted by R_i^a and R_i^{gt}. The centroid of any component C is defined as

$$\mu(C) = \frac{1}{|C|}\sum_{p\in C} p. \tag{27}$$

The predicted and ground-truth centroids for screws and rods are collected as point sets for subsequent matching.

One-to-one matching between predicted and ground-truth components is performed using the Hungarian algorithm. For predicted components P_m and ground-truth components G_n, the matching cost is the Euclidean distance between centroids:

$$d_{mn} = \|\mu(P_m) - \mu(G_n)\|_2. \tag{28}$$

The optimal matching is

$$\pi^* = \arg\min_{\pi} \sum_{m} d_{m,\pi(m)}. \tag{29}$$

If either predicted or ground-truth components are absent, the corresponding sample-level metric is set to invalid for distance-based statistics.

Screw distribution is evaluated by screw count error $E_{\#}$, missed screws N_{miss}, centroid error E_{pos}, Chamfer distance (CD), and the 95th percentile Hausdorff distance (HD95). The screw count error is

$$E_{\#} = |N_{\mathrm{pred}} - N_{\mathrm{gt}}|, \tag{30}$$

and the missed screw count is

$$N_{\mathrm{miss}} = N_{\mathrm{gt}} - N_{\mathrm{match}}, \tag{31}$$

where N_{match} denotes the number of matched screw pairs. The centroid error is computed as the mean Euclidean distance over matched screw pairs:

$$E_{\mathrm{pos}} = \frac{1}{|\mathcal{M}|}\sum_{(p,q)\in\mathcal{M}} \|p - q\|_2, \tag{32}$$

where $\mathcal{M}$ is the set of matched screw centroid pairs. Let $\mathcal{P}_{\mathrm{pred}}$ and $\mathcal{P}_{\mathrm{gt}}$ denote the predicted and ground-truth screw centroid

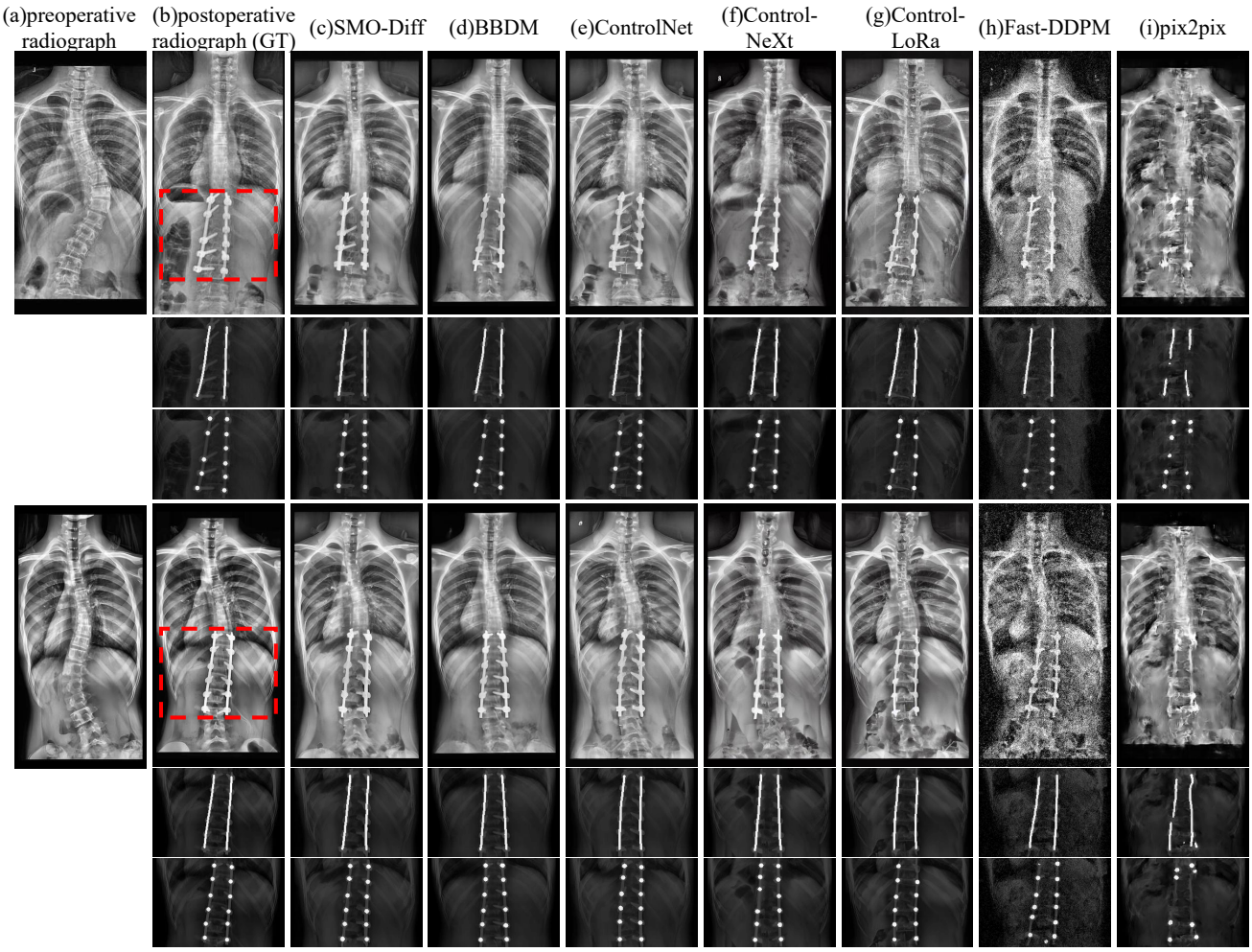


Fig. 5. Visual comparison of postoperative internal-fixation reconstruction across different methods on two representative cases. For each case, the rows show the reconstructed radiograph, rod placement, and pedicle screw distribution from top to bottom. Columns from left to right: preoperative radiograph, postoperative radiograph (GT), SMO-Diff, BBDM [42], ControlNet [21], ControlNeXt [43], ControlLoRA [44], Fast-DDPM [20], and pix2pix [19].

sets. The Chamfer distance is

$$\begin{aligned}\mathrm{CD} = &\frac{1}{2|\mathcal{P}_{\mathrm{pred}}|}\sum_{p\in\mathcal{P}_{\mathrm{pred}}}\min_{q\in\mathcal{P}_{\mathrm{gt}}}\|p-q\|_2 \\ &+\frac{1}{2|\mathcal{P}_{\mathrm{gt}}|}\sum_{q\in\mathcal{P}_{\mathrm{gt}}}\min_{p\in\mathcal{P}_{\mathrm{pred}}}\|q-p\|_2.\end{aligned} \tag{33}$$

The bidirectional nearest-neighbor distance multiset is

$$\begin{aligned}\mathcal{D} = &\left\{\min_{q\in\mathcal{P}_{\mathrm{gt}}}\|p-q\|_2 :\ p\in\mathcal{P}_{\mathrm{pred}}\right\} \\ &\cup\left\{\min_{p\in\mathcal{P}_{\mathrm{pred}}}\|q-p\|_2 :\ q\in\mathcal{P}_{\mathrm{gt}}\right\}.\end{aligned} \tag{34}$$

The screw centroid HD95 is then calculated as the 95th percentile of $\mathcal{D}$:

$$\mathrm{HD95} = P_{95}(\mathcal{D}). \tag{35}$$

For each matched rod pair, let Γ_{pred} and Γ_{gt} denote the skeleton point sets. Rod geometry is evaluated by endpoint error E_{end}, centerline error E_{cline}, and HD95. Since rod endpoints are unordered, let the two endpoints of the predicted rod be e_1, e_2 and those of the ground-truth rod be e_1', e_2'. The endpoint error is computed in a permutation-invariant manner:

$$\begin{aligned}E_{\mathrm{end}} = \min\Big(&\sqrt{\|e_1-e_1'\|_2^2+\|e_2-e_2'\|_2^2}, \\ &\sqrt{\|e_1-e_2'\|_2^2+\|e_2-e_1'\|_2^2}\Big).\end{aligned} \tag{36}$$

The centerline error is the bidirectional average nearest-neighbor distance between Γ_{pred} and Γ_{gt}:

$$\begin{aligned}E_{\mathrm{cline}} = &\frac{1}{2|\Gamma_{\mathrm{pred}}|}\sum_{x\in\Gamma_{\mathrm{pred}}}\min_{y\in\Gamma_{\mathrm{gt}}}\|x-y\|_2 \\ &+\frac{1}{2|\Gamma_{\mathrm{gt}}|}\sum_{y\in\Gamma_{\mathrm{gt}}}\min_{x\in\Gamma_{\mathrm{pred}}}\|y-x\|_2.\end{aligned} \tag{37}$$

Let $\mathcal{D}_{\mathrm{rod}}$ denote the multiset of bidirectional nearest-neighbor distances between Γ_{pred} and Γ_{gt}. The rod centerline HD95 is calculated as the 95th percentile of $\mathcal{D}_{\mathrm{rod}}$:

$$\mathrm{HD95} = P_{95}(\mathcal{D}_{\mathrm{rod}}). \tag{38}$$

For comprehensive ranking, let $r(a,q)$ be the rank of method a on metric q, where a lower value indicates better

TABLE I
COMPARISON OF QUANTITATIVE RESULTS ACROSS DIFFERENT ALGORITHM MODELS. ↓ (↑) INDICATES LOWER (HIGHER) IS BETTER.

Category	Metric	Symbol	SMO-Diff	ControlNet [21]	BBDM [42]	ControlNeXt [43]	ControlLoRA [44]	Fast-DDPM [20]	pix2pix [19]
Rod Geometry Measurement	Endpoint Error ↓	Eend (px)	**9.4**	10.1	10.3	11.8	15.9	12.5	51.5
	Centerline Error ↓	Ecline (px)	**3.9**	4.1	4.4	4.4	6.1	4.1	14.8
	HD95 (Centerline) ↓	HD95 (px)	**7.1**	8	8.1	9.7	15.5	12	57.4
Screw Distribution	Quantity Error ↓	E#	1.3	1.6	**1.1**	2.1	2	2	4
	Missing Screws ↓	Nmiss	**0.4**	0.9	0.6	1.1	1.2	1	3.8
	Centroid Error ↓	Epos (px)	**9.3**	10	10.4	11.3	11.1	10	12.6
	Chamfer Distance ↓	CD (px)	**9.2**	9.7	9.6	10.4	10.4	9.8	12.7
	HD95 (Centroid) ↓	HD95 (px)	**16.3**	17.7	18.5	20.5	20.2	19.6	29.6
Comprehensive Ranking	Rod Ranking Score ↓	Rrod	**1**	2	3.3	4.3	6	4.3	7
	Screw Ranking Score ↓	Rscrew	**1.2**	2.8	2.4	5.6	5.4	3.6	7
	Overall Ranking Score ↓	Roverall	**1.1**	2.4	2.9	5	5.7	4	7
Image Quality	SSIM ↑	–	0.61±0.03	0.59±0.02	**0.64±0.02**	0.63±0.02	0.57±0.02	0.53±0.02	0.61±0.10
	PSNR ↑	(dB)	**16.5±1.4**	15.9±1.2	15.9±1.1	16.0±1.1	15.2±1.2	13.5±1.8	15.4±0.9

TABLE II
QUANTITATIVE RESULTS OF MODULE ABLATION EXPERIMENTS FOR THE SMO-DIFF MODEL. ↓ (↑) INDICATES LOWER (HIGHER) IS BETTER.

Category	Metric	Symbol	SMO-Diff	w/o CMF	w/o Curve Encoder	w/o ROI Encoder
Rod Geometry Measurement	Endpoint Error ↓	Eend (px)	**9.4**	10.3	11.4	55.2
	Centerline Error ↓	Ecline (px)	**3.9**	4.4	4.9	19.7
	HD95 (Centerline) ↓	HD95 (px)	**7.1**	9.2	10.5	70.2
Screw Distribution	Quantity Error ↓	E#	**1.3**	1.9	1.9	6.2
	Missing Screws ↓	Nmiss	**0.4**	1.1	1.3	6.1
	Centroid Error ↓	Epos (px)	**9.3**	10.3	10.1	15
	Chamfer Distance ↓	CD (px)	**9.2**	10.1	10	18.6
	HD95 (Centroid) ↓	HD95 (px)	**16.3**	19	18.9	45.8
Comprehensive Ranking	Rod Ranking Score ↓	Rrod	**1**	2	3	4
	Screw Ranking Score ↓	Rscrew	**1**	2.7	2.3	4
	Overall Ranking Score ↓	Roverall	**1**	2.4	2.7	4
Image Quality	SSIM ↑	–	**0.61 ± 0.03**	0.57 ± 0.02	0.58 ± 0.02	0.57 ± 0.01
	PSNR ↑	dB	**16.50 ± 1.45**	15.91 ± 1.15	15.87 ± 1.21	15.55 ± 0.98

performance. The screw, rod, and overall ranking scores are defined as:

$$R_{\text{screw}} = \frac{1}{5}\big[r(E_{\#}) + r(N_{\text{miss}}) + r(E_{\text{pos}}) + r(\text{CD}) + r(\text{HD95})\big], \tag{39}$$

$$R_{\text{rod}} = \frac{1}{3}\big[r(E_{\text{end}}) + r(E_{\text{cline}}) + r(\text{HD95})\big], \tag{40}$$

$$R_{\text{overall}} = \frac{1}{2}\big(R_{\text{screw}} + R_{\text{rod}}\big). \tag{41}$$

A lower ranking score indicates better overall structural consistency.

In addition, for image quality assessment, the structural similarity index (SSIM [45]) and peak signal-to-noise ratio (PSNR [46]) are also adopted.

C. Controlled Experiment

To validate SMO-Diff for spinal pedicle screw reconstruction, we compare it with ControlNet [21], BBDM [42], ControlNeXt [43], ControlLoRA [44], Fast-DDPM [20], and pix2pix [19]. Quantitative results, including rod geometry, screw distribution, ranking scores, PSNR, and SSIM, are summarized in Table I. Visual comparisons are shown in Fig. 5.

SMO-Diff outperforms all competitors across all quantitative metrics. For rod geometry, it achieves the lowest endpoint error ($E_{\text{end}} = 9.4$ px), centerline error ($E_{\text{cline}} = 3.9$ px), and HD95 (7.1 px), with the highest PSNR (16.50 ± 1.45 dB) and SSIM (0.61 ± 0.03). For screw distribution, it obtains the best $N_{\text{miss}} = 0.4$, $E_{\text{pos}} = 9.3$ px, CD= 9.2 px, and HD95= 16.3 px, ranking first in rod, screw, and overall scores. Though BBDM has the lowest $E_{\#} = 1.1$, it suffers from large

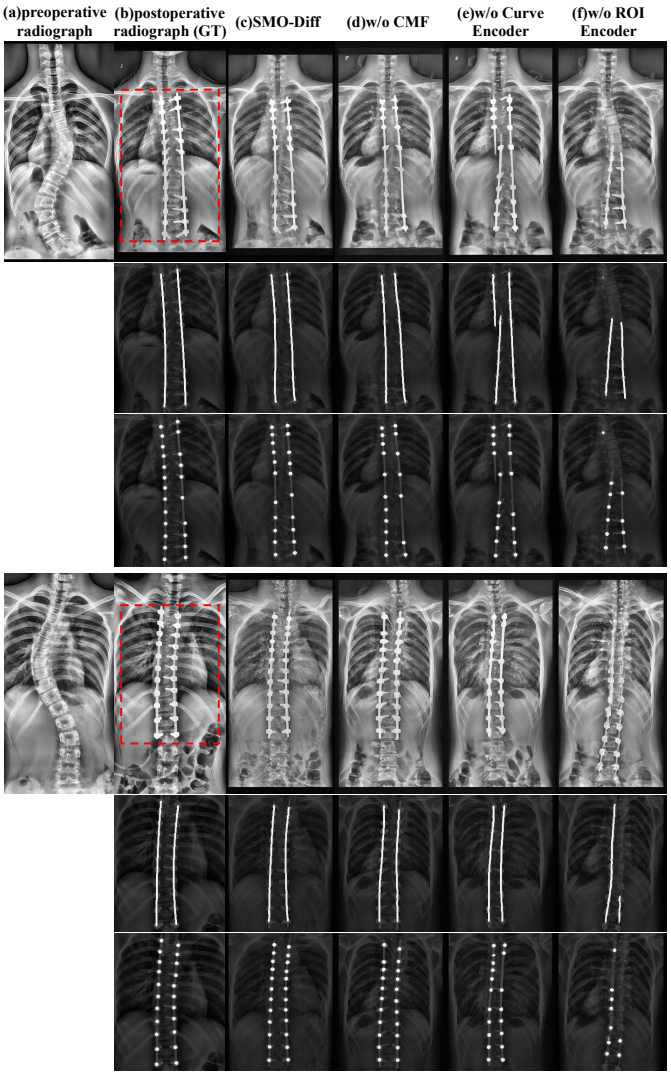


Fig. 6. Ablation study of key components in SMO-Diff on two representative cases. For each case, the rows show full postoperative reconstruction, rod placement, and pedicle screw distribution from top to bottom. The columns show: (a) preoperative radiograph, (b) postoperative radiograph (GT), (c) SMO-Diff, (d) SMO-Diff without CMF, (e) SMO-Diff without Curve Encoder, and (f) SMO-Diff without ROI Encoder.

positional errors. pix2pix performs worst on nearly all metrics, while Fast-DDPM yields noisy and inaccurate reconstructions.

Visual results align with quantitative findings. SMO-Diff generates smooth, continuous rods and precisely aligned screws that closely match the ground truth. ControlNet and BBDM produce blurry edges and minor misalignments. ControlNeXt and ControlLoRA show inconsistent rod curvature and uneven screw spacing. Fast-DDPM introduces severe noise, broken rods, and incomplete screw patterns. pix2pix exhibits the worst performance, with distorted implants, discontinuous rods, and scattered screw distribution. Overall, SMO-Diff achieves superior geometric accuracy, distribution completeness, and visual fidelity.

D. Ablation Experiment

To validate the contribution of each key component in SMO-Diff, we conduct ablation studies by removing the CMF module, Curve Encoder, and ROI Encoder individually. Quantitative results are summarized in Table II, including rod/screw metrics and image quality (PSNR/SSIM), with visual comparisons provided in Fig. 6.

The full SMO-Diff achieves optimal performance across all metrics, obtaining the highest PSNR (16.50 ± 1.45 dB) and SSIM (0.61 ± 0.03). Removing CMF causes moderate degradation: rod endpoint error increases to 10.3 px, missing screws rise to 1.1, and PSNR/SSIM drop to 15.91 ± 1.15 dB / 0.57 ± 0.02. Removing the Curve Encoder further impairs geometric consistency, with centerline error of 4.9 px and PSNR/SSIM of 15.87 ± 1.21 dB / 0.58 ± 0.02. Excluding the ROI Encoder leads to severe performance collapse: endpoint error surges to 55.2 px, missing screws reach 6.1, and PSNR/SSIM decline to 15.55 ± 0.98 dB / 0.57 ± 0.01.

Visual results in Fig. 6 are consistent with Table II: the full model yields smooth rods and precise screw distributions matching the ground truth. Removing CMF introduces minor blurriness; removing the Curve Encoder causes distorted rod shapes; without the ROI Encoder, the model fails to localize the spinal region, resulting in invalid reconstructions. In summary, the ROI Encoder, Curve Encoder, and CMF are all indispensable for accurate spinal reconstruction.

V. Discussion

A. Manual Centerline Annotation Limitations and Ongoing Automatic Detection Study

In the current SMO-Diff workflow the spinal centerline and surgical ROI required as model inputs rely entirely on manual labeling conducted by experienced spinal surgeons. Double-blind manual annotation ensures high-quality labels for model training and evaluation. However manual marking consumes substantial clinical time and human resources. Annotation outcomes are susceptible to individual experience differences among clinicians. These issues hinder convenient clinical deployment of the proposed model. Our research team has launched additional research targeting automatic extraction of vertebral landmarks and spinal centerline contours. This follow-up work is designed to eliminate manual labeling dependence for centerline map generation. Once finished it can be seamlessly integrated into the front end of SMO-Diff. Such integration will build a fully automatic prediction pipeline from raw preoperative radiographs to postoperative imaging and implant prediction and greatly improve the practical application value of this preoperative planning tool.

B. Functional and Clinical Values of Three Core Modules

The superior performance of SMO-Diff can be attributed to the complementary roles of its three key modules. First, the Curve Encoder extracts global spinal alignment and curvature information from the predicted postoperative centerline, providing a morphology-aware prior that guides anatomically plausible correction patterns. Second, the ROI Encoder focuses

on the surgically relevant region and supplies localized spatial constraints associated with implant placement and postoperative tissue deformation. This local prior is particularly important for accurately reconstructing screw distribution and rod geometry. Third, the proposed CMF module explicitly models the interaction between global spinal morphology and local surgical-region information through bidirectional cross-attention. By integrating these complementary features, CMF enables the diffusion network to simultaneously preserve global structural consistency and local fixation accuracy. The ablation results demonstrate that removing any of these modules leads to performance degradation, confirming that both global morphology guidance and local surgical constraints are essential for reliable postoperative outcome prediction.

C. Limitations and Future Work

Despite the promising quantitative and visualized experimental results achieved by SMO-Diff, several limitations remain to be improved in future explorations. First, the dataset used for model training is sourced from a single hospital with limited patient population and relatively unified surgical treatment habits, and further multi-center external validation based on multi-institutional heterogeneous AIS patient data is needed to verify the generalization ability of the model across different hospital diagnosis and treatment standards. Second, preoperative spine morphology maps and surgical ROI masks still rely on manual landmark annotation by experienced surgeons, which increases preprocessing workload and hinders fully automatic pipeline deployment. Third, the current model only completes two-dimensional postoperative frontal radiographic prediction, lacking the capability to generate 3D spinal reconstruction results and three-dimensional screw placement simulation which is more valuable for clinical biomechanical evaluation. In future work, we plan to develop an automatic vertebral keypoint detection algorithm to generate spine morphology maps without manual labeling, expand the dataset with paired biplanar anteroposterior and lateral spinal radiographs, extend SMO-Diff framework toward 3D conditional diffusion generation, and combine intraoperative biomechanical parameters to further optimize the prediction accuracy of correction rate and implant layout, promoting the transformation of this algorithm into practical clinical auxiliary planning software.

VI. Conclusion

We proposed SMO-Diff, a spinal morphology-oriented conditional diffusion framework for postoperative image generation and internal-fixation layout prediction in scoliosis surgery. By treating the postoperative spinal centerline as a global morphological constraint, designing the Curve Encoder and ROI Encoder for global curve and local surgical-region features, and introducing CMF for multi-source feature fusion, SMO-Diff generates anatomically consistent and clinically plausible postoperative radiographs. The predicted screw and rod masks further provide key surgical planning information, offering a visual and quantitative basis for fusion-level assessment and related planning parameters. These results suggest that SMO-Diff can provide reliable technical support for intelligent preoperative planning in scoliosis surgery.

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